

AI Travel Concierge Platform

◆◆◆ Technical Specification v6.2

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AI Travel Concierge Platform

Technical Specification

Version 6.2 — Pre-Development Hardening Applied

Version history:

- **v6.2 (2026-05-17):** Pre-development hardening pass. ~50 spec edits across 7 batches addressing all Critical and 17 of 18 High red-team findings, plus 9 curated Medium items and one operator-designed feature (authenticated customer chat rate limit, §24.9). See `Merge_Notes.md` for edit-by-edit log; see `Memory.md` D-022 through D-040 for design decisions and rejected alternatives.
- **v6.1:** Consolidated with Self-Service Help.

Sections substantially changed in v6.2 (not exhaustive — see merge notes):

- §1.5, §5.1, §5.4 (new), §10.1, §10.5a (new), §11.2, §11.5, §11.6, §13.5, §14.0 (new), §14.3, §14.4, §14.7, §15.7, §15.14, §16.3, §16.6, §16.7, §17.10, §18.5, §18.9, §19.3, §21.10.1 (new), §22.4a (new), §24.5, §24.8, §24.9 (new), §24.10 (renumbered), §25.4a (new), §26.2, §26.3, §26.3a (new), §26.5a (new), §28.13, §28.16, §30.8

Operator-confirm markers remaining in v6.2 (require attorney engagement or operator action before Phase 2):

- §15.7 — Seller of Travel attorney engagement
- §15.14.6 — ToU/ICA chunk-license-survival language
- §16.7.1 — Legal-page attribution wording
- §25.4a — Tenant-owned-notes treatment after CCPA deletion

Sections 1-32 + Appendices A and B

Complete v6.1 specification

Generated: 2026-05-16

About This Document

This is the consolidated v6.1 technical specification for a multi-tenant AI-powered cruise travel platform. It captures the full architecture, schema, business logic, and operational design across three business prongs: (1) a fully-automated independent travel agency, (2) a BYO-host SaaS for independent travel agents, and (3) a white-label sub-host model where independent agents operate under the platform's host agency relationship.

The document supersedes prior drafts (v4.0, the v5.x series of addendums, the v6.0 consolidated spec, and the Self-Service Help Addendum v1.0) and incorporates all design decisions made during the consolidated drafting process.

Status

All 32 sections plus Appendix A (Fresh-Build Notes) and Appendix B (Glossary) are present in this document. Open items pending resolution are tracked in `MEMORY.md`, the working decision log.

Audience

Primary: the platform operator (technical product owner) and the engineering team that will implement the system. Secondary: legal, finance, and operational stakeholders who need to understand the system's behavior and constraints.

Conventions

- Schema is presented in PostgreSQL syntax with Supabase-flavored Row Level Security (RLS) policies.
- Code examples use TypeScript unless otherwise noted.
- ‘Tenant’ refers to any account operating on the platform: BYO-host tenants, sub-host tenants, or the platform’s own automated agency.
- ‘Customer’ refers to the end traveler being served by a tenant.
- Sections end with ‘Calls Worth Flagging’ — design decisions or trade-offs that may need revisiting.
- Items marked ‘TBD’ or ‘Deferred’ are intentional gaps to resolve in implementation or in future versions.

Section 1 – Platform Overview

The platform delivers AI-powered cruise travel agency services across three business models, sharing a common technical foundation. This section defines the business model, the personas the platform serves, and the high-level technical posture.

1.1 Business Model — Three Prongs

Prong 1: Fully Automated Own Agency

The platform operator runs their own cruise travel agency. AI personas handle customer inquiries, recommend trips, generate quotes, and submit bookings. Commissions accrue to the platform directly. This is the platform's first-party revenue stream.

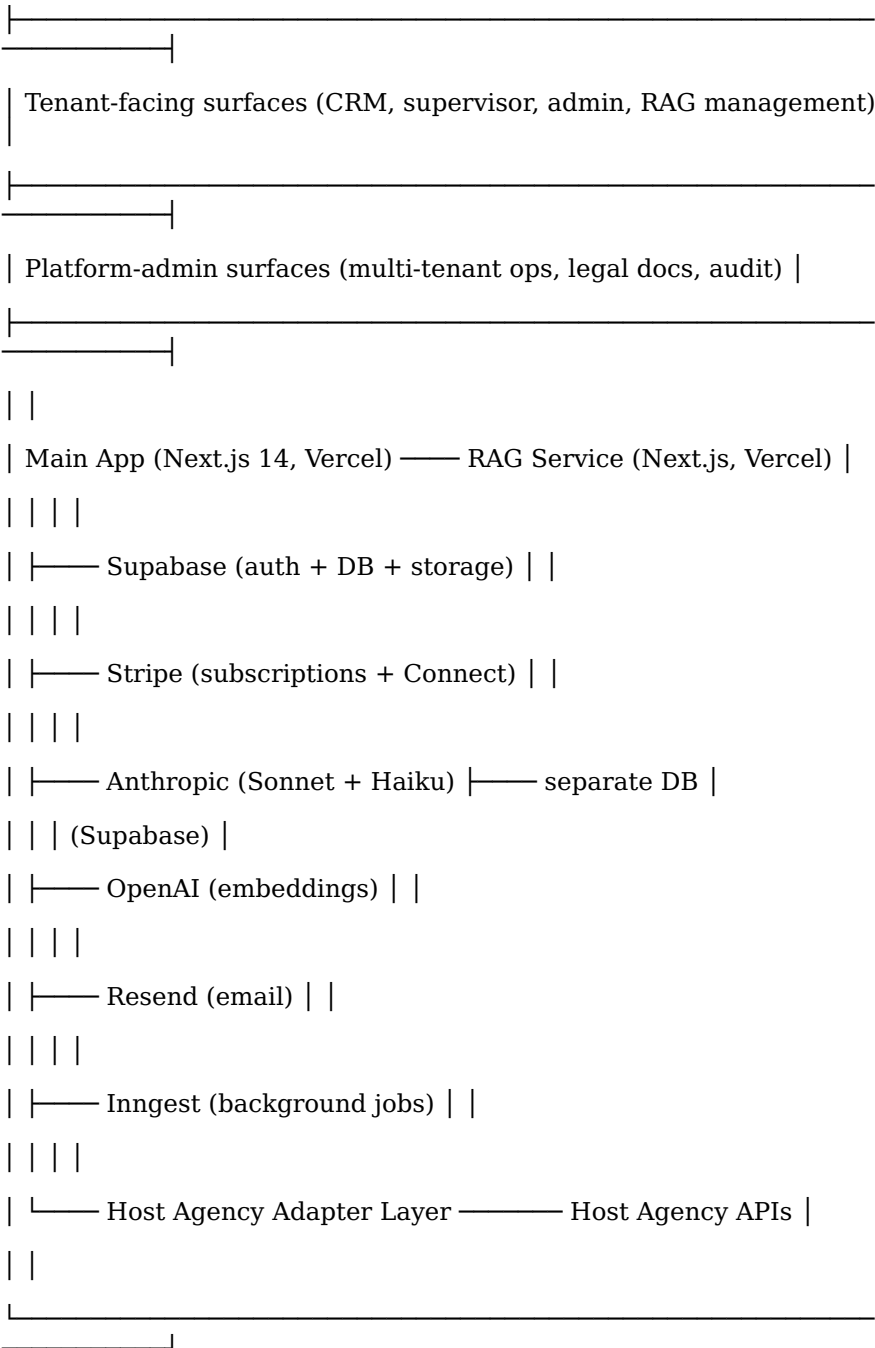
Prong 2: BYO-Host SaaS for Independent Agents

Independent travel agents who already have their own host agency relationship (e.g., with Travel Leaders, Avoya, Cruise Planners) subscribe to the platform as software. They bring their own host agency credentials. The platform charges a per-seat subscription; commissions flow to their existing host agency arrangements without platform intermediation.

Prong 3: White-Label Sub-Host Model

Independent travel agents who don't have their own host agency relationship subscribe to the platform AND book under the platform's host agency. The platform takes a commission split (with tiered rates) and remits the remainder via Stripe Connect Express.

1.2 Technical Posture at a Glance



1.3 Personas Served

Persona	Description	Primary Surfaces
Customer	End traveler being served by a tenant agency	Chat, quote view, booking flow, group invite landing
Tenant Owner	Operator of a sub-host or BYO-host tenant	Tenant admin, CRM, supervisor, settings
Tenant Agent	Sub-user of a tenant; assists with	CRM, chat history, quote builder,

	customers	booking flow
Tenant Billing Admin	Manages subscription and payouts	Billing UI, Stripe Connect dashboard
Platform Super Admin	Operates the platform; full access	Admin console, RAG global review, tenant ops
Platform Compliance	Handles tenant reviews and legal	Admin console, onboarding queue, legal docs
Platform Finance	Reconciles commissions, payouts	Admin console, finance dashboards
Platform Support	Customer-facing support across tenants	Read-only admin views, ticket integration
AI Persona	The AI assistant the customer interacts with	Chat, drafted emails, suggested quotes
Group Coordinator	Customer organizing a group booking	Coordinator portal, forum chat, RSVPs
Group Invitee	Customer invited to a group	Invitation landing, forum chat, RSVP

1.4 Tenant Resolution

Every request to the platform resolves to a tenant context based on the request's host:

- Platform primary domain (e.g., ai-travelconciierge.com) → platform context (admin operations, signup landing)
- Tenant subdomain on platform (e.g., smithtravel.ai-travelconciierge.com) → that tenant
- Tenant custom domain (e.g., travel.smithfamilyagency.com) → that tenant via DNS CNAME (per §16.3)

Tenant resolution happens in middleware and is cached for the request lifecycle. RLS policies enforce tenant scoping at the database level.

1.5 Multi-Tenancy Architecture

The platform enforces tenant isolation through four layers (§26.2): identity (Supabase Auth), tenant context (middleware tenant resolution), authorization (assertPermission()), and data access scoping (tenantClient(ctx) plus RLS).

All tenant-scoped tables include a tenant_id column with RLS policies enforcing access. Two helper functions are used in policies: auth_user_in_tenant(tenant_id) checks the user has an active row in that tenant; tenant_is_active(tenant_id) checks the tenant itself is operational. See §5.1 for definitions and §5.1.2 for required policy coverage.

Database access from application code goes through typed clients per §5.4. Service-role usage discipline is defined in §26.3a.

The platform supports a strictly two-level structure: the platform, and direct tenants (BYO-host, sub-host, or the platform’s own automated agency). No tenant nests beneath another tenant. If a sub-host engages subcontractors to help operate their agency, those subcontractors are users inside the sub-host’s tenant — not separate tenants on the platform. The platform has no relationship with a sub-host’s subcontractors.

Section 2 — Technology Stack

Concrete technology choices, with rationale and scope.

2.1 Application Runtime

Component	Choice	Notes
Frontend framework	Next.js 14 (App Router)	Server components, SSR, streaming
Hosting	Vercel	Edge functions, automatic TLS, deployment
Backend language	TypeScript (Node.js runtime)	Shared types between frontend and backend
Mobile	Responsive web only (no native apps in v6)	Design preserves future native option
UI library	shadcn/ui + Tailwind CSS	Component composition; tenant theming via CSS vars

2.2 Data & State

Component	Choice	Notes
Database	Supabase (PostgreSQL)	Main app DB; separate Supabase project for RAG
Auth	Supabase Auth	OAuth providers: Google, Microsoft, Facebook
File storage	Supabase Storage	Logos, generated images, statement PDFs
Row-level security	Postgres RLS	Tenant scoping enforced at DB layer
Vector embeddings	pgvector in RAG Supabase	OpenAI text-embedding-3-small (1536-dim)
Job queue / cron	Inngest	Background tasks, scheduled jobs, retries

2.3 AI Services

Service	Use	Notes
Claude Sonnet (Anthropic)	Conversational chat, drafting, complex reasoning	Streaming via SSE
Claude Haiku (Anthropic)	Classification, screening, entity extraction, lightweight tasks	Cost-effective for high-volume calls
OpenAI Embeddings	Vector embeddings for RAG	text-embedding-3-small @ 1536 dimensions
Image generation	Replicate (Stable Diffusion XL) or OpenAI DALL-E 3	For group hero images, destination imagery; choice deferred to implementation

2.4 Payments

Component	Choice	Notes
Subscription billing	Stripe Subscriptions	Per-tenant subscriptions for SaaS access
Marketplace payouts	Stripe Connect Express	Sub-host KYC + payouts via standard Connect flow
Tax reporting	Stripe Connect (1099-NEC automatic)	Stripe files 1099s for sub-hosts $\geq$ \$600/year
Currency	USD only at launch	Architecture supports adding currencies later

2.5 Communication

Channel	Provider	Notes
Outbound email	Resend	React Email templates, tenant-from domain support
Inbound email (Gmail)	Gmail API via Google Pub/Sub	Optional per-tenant Gmail integration
SMS (Phase 3)	Twilio (planned)	Out of scope for v6 launch
Webhooks	Stripe, Resend, Gmail Pub/Sub	Signature verification mandatory

2.6 Observability

- suspended — read-only; no new bookings; existing commissions continue to settle
- terminated — closed; data anonymized per retention policy

3.3 Tier Matrix

Tiers determine pricing, feature gates, commission split (sub-hosts only), and quota limits.

BYO-Host Tiers

Tier	Monthly	Annual	Seats	Description
BYO Research	\$19	\$190	1	Single-agent, exploration / research use
BYO Professional	\$59	\$590	1	Single-agent professional with full CRM
BYO Agency	\$99 base (see ladder)	\$990 base (see ladder)	1 included; tiered add-on	Multi-agent agency with custom domain

BYO Agency seat ladder (additional users beyond the included first seat):

Seat range	Monthly per seat	Annual per seat
User 1 (included in base)	\$99	\$990
Users 2-4	\$59	\$590
Users 5-10	\$49	\$490
Users 11+	\$39	\$390

Annual = 10× monthly (two months free) applied at every tier of the ladder. Example: an Agency tenant with 6 users pays \$99 + (3 × \$59) + (2 × \$49) = \$374/mo, or \$3,740/yr.

Sub-Host Tiers

Sub-host tiers differ on commission split (platform’s share / sub-host’s share), persona customization, custom domain access, and support level.

Tier	Monthly	Commission Split	Persona Customization	Custom Domain?
Sub-Host Starter	\$49	30 / 70	None	No
Sub-Host Pro	\$149	20 / 80	Display name rename	No
Sub-Host Agency	\$249 base (see ladder)	10 / 90	Display name + prompt addendum	Yes

Sub-Host Agency seat ladder (additional users beyond the included first seat; same rates as BYO Agency):

Seat range	Monthly per seat	Annual per seat
User 1 (included in base)	\$249	\$2,490
Users 2-4	\$59	\$590
Users 5-10	\$49	\$490
Users 11+	\$39	\$390

Annual = 10× monthly (two months free) applied at every tier of the ladder. Starter and Pro are single-seat only.

NOTE:** Agency tier adds a Haiku-screened prompt addendum (per §16.6), custom domain support, dedicated success manager, and multi-seat support. Both Pro and Agency get persona display-name renaming.

3.4 Hold Periods

Default hold periods between commission receipt and payout availability per §14. Tenant-configurable; minimums apply by tier.

Tier	Default Hold (Days)	Minimum
Sub-Host Starter	7	7 (cannot go lower)
Sub-Host Pro	3	0
Sub-Host Agency	0 (immediate)	0

3.5 Trial Period

All paid tenant tiers get a 30-day trial. The trial starts at activation (not at signup), so a sub-host whose review takes a week still gets full 30 days when they’re approved.

3.6 Tenant Resolution Logic

```
// apps/main/src/middleware.ts (excerpt)

async function resolveTenant(req: NextRequest): Promise<Tenant | null> {

  const hostname = req.headers.get('host');

  // Platform primary domain → platform context

  if (hostname === PLATFORM_PRIMARY_DOMAIN) return null;

  // Subdomain on platform

  const subdomainMatch = hostname.match(new RegExp(`^([a-z0-9-]+).${PLATFORMDOMAINREGEX}`));

  if (subdomainMatch) {

    return await getTenantBySlug(subdomainMatch[1]);

  }

}
```

```
// Custom domain

return await getTenantByCustomDomain(hostname);

}
```

3.7 Multi-Tenant Identity

A single OAuth account (one auth.users row) can be associated with multiple public.users rows across different tenants. The user has a separate identity, memory, and CRM context in each tenant. UI offers a tenant context switcher when applicable.

Section 4 — Detailed Feature Matrix

Per-feature gating across all tiers. ‘Y’ = included; ‘N’ = not available; specific values where indicated.

Feature	BYO Research	BYO Pro	BYO Agency	Sub Starter	Sub Pro	Sub Agency
AI Chat (customer-facing)	Y	Y	Y	Y	Y	Y
AI Personas (all 6)	Y	Y	Y	Y	Y	Y
CRM with contacts	Y	Y	Y	Y	Y	Y
Quote generation	Y	Y	Y	Y	Y	Y
Booking flow	Y	Y	Y	Y	Y	Y
Memory & rapport	Y	Y	Y	Y	Y	Y
RAG knowledge base	Y	Y	Y	Y	Y	Y
Group bookings	N	Y	Y	Y	Y	Y
Forum chat	N	Y	Y	Y	Y	Y
Gmail integration	N	Y	Y	Y	Y	Y
Supervisor dashboard	N	Y	Y	Y	Y	Y
Persona renaming	N	N	Y	N	Y	Y
Persona prompt addendum	N	N	N	N	N	Y
Custom domain	N	N	Y	N	N	Y